

INDIAN GEOGRAPHY

Physical, Economic & Human Geography

For SSC | UPSC | Railway | Banking | State PSC | Defence Exams

1. INDIA - BASIC FACTS

- > Area: 32.87 lakh sq km (7th largest). 2.4% of world's land area.
- > Location: 8'4"N to 37'6"N latitude; 68'7"E to 97'25"E longitude.
- > Tropic of Cancer (23.5'N) passes through 8 states: Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram.
- > Standard Meridian: 82'30"E (passes through Mirzapur, UP). IST = GMT + 5:30.
- > Total states: 28. Union Territories: 8. Total coastline: 7,516 km.
- > Land borders: 15,200 km. Neighbours: Pakistan, China, Nepal, Bhutan, Bangladesh, Myanmar, Sri Lanka (sea), Maldives (sea).
- > Longest border with: Bangladesh (4,096 km). Shortest: Afghanistan (106 km, via PoK).
- > Northernmost point: Indira Col (Siachen). Southernmost: Indira Point (Great Nicobar). Southernmost mainland: Kanyakumari.

2. PHYSICAL FEATURES

The Himalayas:

- > Young fold mountains. Three parallel ranges: Greater/Himadri (highest), Lesser/Himachal (middle), Shiwalik (outer/lowest).
- > Highest peak: Mt. Everest (8,849m, Nepal). India's highest: Kanchenjunga (8,586m, Sikkim).
- > K2/Godwin Austen (8,611m) in PoK - 2nd highest in world.
- > Important passes: Karakoram (Ladakh, highest), Rohtang (HP), Nathu La & Jelep La (Sikkim), Bomdi La (Arunachal).

Northern Plains:

- > Alluvial deposits by Indus, Ganga, Brahmaputra. Bhabar > Terai > Bhangar (old alluvium) > Khadar (new alluvium).
- > Most fertile agricultural region. Very dense population.

Peninsular Plateau:

- > Oldest landmass (part of Gondwanaland). Deccan Plateau, Central Highlands, Western & Eastern Ghats.
- > Western Ghats: Continuous, higher (avg 1200m). Eastern Ghats: Discontinuous, lower.
- > Highest peak of Western Ghats: Anamudi (2,695m, Kerala). Eastern Ghats: Mahendragiri (1,501m).

Coastal Plains & Islands:

- > Western coast: Konkan (Maharashtra), Malabar (Kerala). Eastern coast: Coromandel (TN), Northern Circars (AP).
- > Andaman & Nicobar: 572 islands, Bay of Bengal. Lakshadweep: 36 islands, Arabian Sea (coral).

3. DRAINAGE SYSTEM

Rivers flowing into Bay of Bengal (East):

- > Ganga: Longest river system in India (2,525 km). Origin: Gangotri glacier (Uttarakhand). Tributaries: Yamuna, Son, Ghaghara, Gandak, Kosi.
- > Brahmaputra: Origin in Tibet (Tsangpo). Enters India in Arunachal (Dihang). Largest river island: Majuli (Assam).
- > Godavari: Longest peninsular river (1,465 km). 'Dakshin Ganga'. Origin: Nashik (Maharashtra).
- > Krishna: 2nd longest peninsular (1,400 km). Origin: Mahabaleshwar (Maharashtra).
- > Kaveri: Origin: Talakaveri, Kodagu (Karnataka). Shivanasamudra Falls. Largest delta in South India.
- > Mahanadi, Damodar (Sorrow of Bengal), Subarnarekha.

Rivers flowing into Arabian Sea (West):

- > Indus: Origin in Tibet (Mansarovar). Length in India: 1,114 km. Tributaries: Jhelum, Chenab, Ravi, Beas, Sutlej.
- > Narmada: Origin Amarkantak (MP). Flows through a rift valley. Marble Rocks at Jabalpur.
- > Tapi/Tapti: Origin Multai (MP). Also flows through rift valley.
- > Sabarmati, Mahi, Luni (only river in Rajasthan desert, inland drainage).

Important Lakes:

- > Largest saltwater lake: Chilika (Odisha). Largest freshwater: Wular (J&K).
- > Highest lake: Pangong Tso (Ladakh). Dal Lake (Srinagar), Loktak (Manipur - floating island Keibul Lamjao NP).

4. CLIMATE

- > India has tropical monsoon climate. 4 seasons: Winter (Dec-Feb), Summer (Mar-May), SW Monsoon (Jun-Sep), NE Monsoon (Oct-Nov).
- > SW Monsoon brings 75% of India's rainfall. Arrives Kerala coast by June 1.
- > Two branches: Arabian Sea branch (western India) and Bay of Bengal branch (eastern India).
- > Mawsynram (Meghalaya): Wettest place on earth. Cherrapunji: 2nd wettest.
- > Driest place: Jaisalmer (Rajasthan). Coldest inhabited: Dras (Ladakh).
- > NE Monsoon: Brings rain to Tamil Nadu coast (Oct-Dec). Chennai gets most rain in winter.
- > El Nino: Warming of Pacific Ocean. Causes drought/less monsoon in India.
- > La Nina: Cooling of Pacific. Causes heavy monsoon/floods in India.
- > Indian Ocean Dipole (IOD): Affects monsoon patterns.

5. SOILS OF INDIA

- > Alluvial Soil: Most widespread (40% area). In Indo-Gangetic plain. Most fertile. Good for rice, wheat, sugarcane.
- > Black Soil (Regur): Deccan Plateau (Maharashtra, Gujarat, MP). Made from basalt/lava. Good for cotton ('Black Cotton Soil').
- > Red Soil: Deccan Plateau (eastern/southern). Iron oxide gives red color. Porous, less fertile.
- > Laterite Soil: Heavy rainfall areas (Kerala, Karnataka, NE). Leached soil. Good for tea, coffee, cashew.
- > Desert/Arid Soil: Rajasthan, Gujarat. Sandy, low moisture. High salt content.
- > Mountain/Forest Soil: Himalayan region. Rich in humus at lower altitude.
- > Peaty/Marshy Soil: Kerala, Bihar, Uttarakhand. Waterlogged, acidic, heavy organic matter.

6. AGRICULTURE

- > India: 2nd largest agricultural producer globally. ~42% workforce in agriculture.
- > Green Revolution (1960s): HYV seeds, irrigation, fertilizers. Led by MS Swaminathan. Focus: wheat & rice.
- > White Revolution (Operation Flood): Milk production. Verghese Kurien. India: largest milk producer.
- > Blue Revolution: Fish/aquaculture production. Yellow Revolution: Oilseeds. Golden Revolution: Horticulture/Honey.

Major Crops:

- > Rice: Kharif crop. Largest producer: West Bengal. India is 2nd globally (after China). NOW India is #1 (2026).
- > Wheat: Rabi crop. Largest: UP, then Punjab, Haryana, MP.
- > Sugarcane: UP (largest), Maharashtra, Karnataka.
- > Cotton: Maharashtra, Gujarat, Telangana. 'White Gold'.
- > Tea: Assam (largest), West Bengal (Darjeeling), Tamil Nadu (Nilgiris).
- > Coffee: Karnataka (Coorg - 70% of India's coffee), Kerala, Tamil Nadu.
- > Rubber: Kerala (90% of India's rubber). Natural rubber - tropical crop.
- > Spices: Kerala (Spice Garden of India). Pepper, cardamom, cloves.

Crop Seasons:

- > Kharif (June-Oct): Rice, Maize, Jowar, Bajra, Cotton, Jute, Groundnut, Soybean.
- > Rabi (Oct-March): Wheat, Barley, Gram, Mustard, Peas, Linseed.
- > Zaid (March-June): Watermelon, Muskmelon, Cucumber, Vegetables.

7. MINERALS & ENERGY RESOURCES

- > Iron Ore: Jharkhand, Odisha, Chhattisgarh, Karnataka, Goa. India: 4th largest producer.
- > Coal: Jharkhand (Jharia - largest coalfield), Odisha, Chhattisgarh, WB (Raniganj). India: 2nd largest producer.
- > Petroleum: Mumbai High (offshore - largest), Assam (Digboi - oldest), Gujarat (Ankleshwar), Rajasthan (Barmer).
- > Natural Gas: Krishna-Godavari Basin (largest), Mumbai High, Gulf of Cambay.
- > Bauxite (Aluminium ore): Odisha (largest), Gujarat, Jharkhand, Maharashtra.
- > Mica: Jharkhand, Bihar, Andhra Pradesh, Rajasthan. India: largest producer.
- > Manganese: Odisha, Karnataka, MP. Used in steel manufacturing.
- > Gold: Karnataka (Kolar Gold Fields - now closed), Rajasthan (Hutti).
- > Nuclear Minerals: Uranium (Jaduguda, Jharkhand), Thorium (Kerala - monazite sands, world's largest reserve).

Power Generation:

- > Thermal Power: 60%+ of total. Coal-based mainly.
- > Renewable: Solar (Rajasthan, Gujarat largest), Wind (Tamil Nadu, Gujarat, Karnataka), Hydro.
- > Nuclear: Tarapur (Maharashtra - first), Kalpakkam (TN), Kudankulam (TN - largest).

8. INDUSTRIES & TRANSPORT

Major Industrial Regions:

- > Mumbai-Pune: Cotton textiles, petrochemicals, engineering.
- > Hooghly (Kolkata): Jute, engineering, chemicals.
- > Bangalore: IT capital, aerospace (HAL), electronics.
- > Ahmedabad: Textiles (Manchester of India), chemicals.
- > Chennai-Coimbatore: Automobiles (Detroit of India), textiles.

Transport:

- > Railways: 4th largest network globally. ~68,000 km route. Largest employer. 100% electrification target achieved (99.2% by Nov 2025).
- > First railway: Mumbai to Thane (1853, 34 km). Longest route: Dibrugarh to Kanyakumari.
- > National Highways: Total length ~1.46 lakh km. NH-44 (longest: Srinagar to Kanyakumari).
- > Golden Quadrilateral: Connects Delhi-Mumbai-Chennai-Kolkata (5,846 km).
- > Major Ports: 12 major ports. Largest by cargo: Deendayal (Kandla), Gujarat. Newest: Vadhavan (Maharashtra).
- > Airports: Busiest: Delhi (IGI Airport). Largest: Rajiv Gandhi International (Hyderabad by area).